Chapter 9: Cellular Respiration and Fermentation

Complete Master Reference Edition • Fused Curriculum (All 6 Modules, Lecture Slides & Practice Questions)

Course: BSC2010C General Biology I

Institution: University of Central Florida

Source: Canvas Modules, Instructor Slides & Active Learning #3

Learning Objectives

- Understand the role of electron carriers in Cellular Respiration
- Describe the process of Glycolysis including energy investment and energy payoff
- Explain how pyruvate oxidation connects Glycolysis with the Citric Acid Cycle
- Relate the Citric Acid Cycle to carbon and electron flow
- Describe the structure and function of the electron transport chain (ETC)
- Explain how the hydrogen ion concentration gradient connects the ETC with ATP production
- Compare substrate level phosphorylation with oxidative phosphorylation

Chapter 9: Cellular Respiration and Fermentation

Chapter 9 continues the energy conversation by covering the many stages and processes of Cellular Respiration and Fermentation - how cells convert the chemical energy in food to ATP. Before you can understand how cells do this, there are two concepts that you must understand first: Redox Reactions and Phosphorylation.

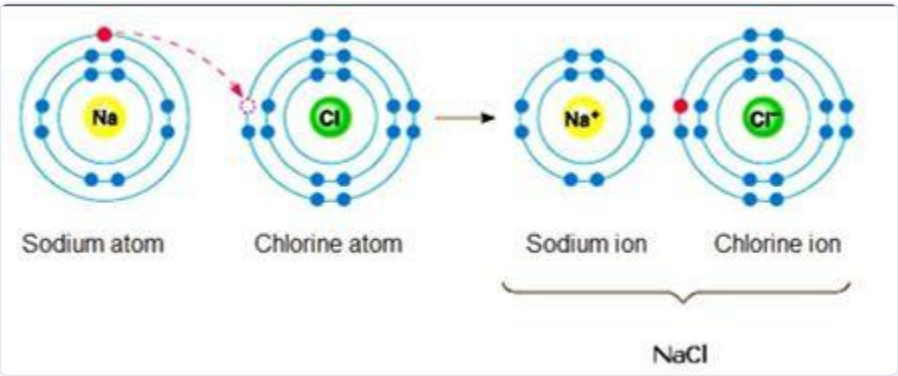
The chemical energy contained within the bonds and electrons of the food we eat are used to add a phosphate to ADP making it ATP - the energy currency of the cell. Reduction Oxidation Reactions, also called Redox reactions, transfer electrons from one substance to another. In a Reduction reaction a substance gains electrons, and in an Oxidation reaction a substance loses electrons. An easy way to remember this is LEO-GER:



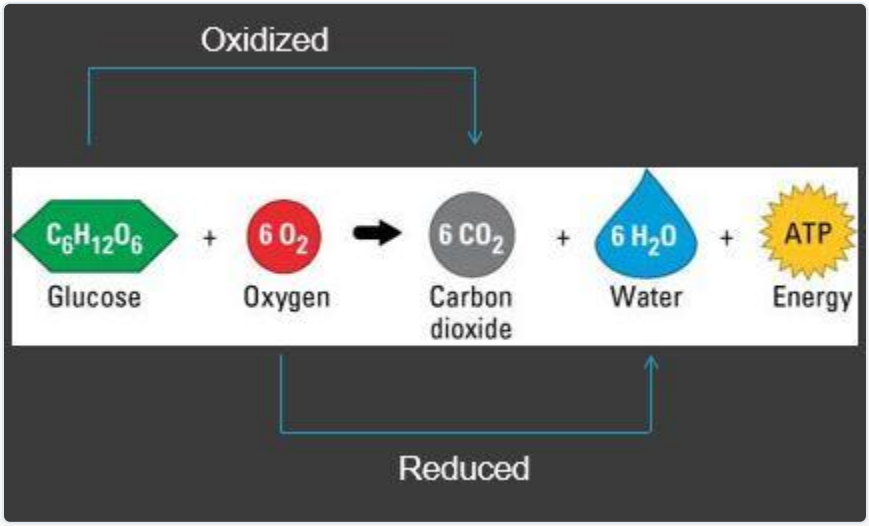
LEO = **L**ose **E**lectrons, **O**xidized

GER = **G**ain **E**lectrons, **R**educed

A familiar example of a Redox Reaction is the reaction that occurs to create the ionic bond in NaCl: Sodium loses an electron and is oxidized. Chlorine gains an electron and is reduced.

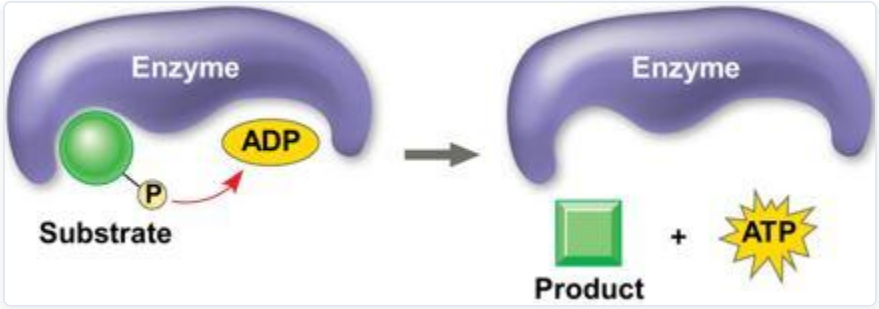


Cellular Respiration is a Redox Reaction. In Cellular Respiration, cells convert glucose and oxygen into carbon dioxide, water, and ATP. Glucose loses electrons and is oxidized to carbon dioxide. Oxygen gains electrons and is reduced to water.



Interactive Course Module: Redox Reaction widget

To make ATP, cells add a phosphate to ADP to make ATP, this process is called phosphorylation. There are two types of phosphorylation that occur during cellular respiration: Substrate Level Phosphorylation and Oxidative Phosphorylation. Substrate Level Phosphorylation is enzyme catalyzed; a substrate with a phosphate and an ADP bind to the active site of an enzyme, the enzyme catalyzes the removal of the phosphate from the substrate and adds it to the ADP to make ATP.



In Oxidative Phosphorylation ATP is generated by the addition of a free (i.e., unbound) inorganic phosphate to ADP. This is powered by the electron transport chain in cellular respiration.

Interactive Course Module: Phosphorylation widget

CHAPTER READING

Chapter 9 Textbook reading Energy and ATP, Glycolysis, Pyruvate Oxidation, Citric Acid Cycle, Oxidative Phosphorylation

CHAPTER LINKS

Chapter 9 PowerPoint: Part 1 ; Part 2

Chapter 9 Post-Quiz

Chapter 9 Simple Practice Problems

Section 9.1: ATP and Energy in Living Systems

Electrons and energy transfer, NAD^+/NADH dinucleotide structure, ATP phosphoanhydride architecture, and substrate-level phosphorylation.

Learning Objectives

By the end of this section, you will be able to do the following:

- Discuss the importance of electrons in the transfer of energy in living systems
- Explain how ATP is used by cells as an energy source

Energy production within a cell involves many coordinated chemical pathways. Most of these pathways are combinations of oxidation and reduction reactions, which occur at the same time. An oxidation reaction strips an electron from an atom in a compound, and the addition of this electron to another compound is a reduction reaction. Because oxidation and reduction usually occur together, these pairs of reactions are called oxidation reduction reactions, or redox reactions.

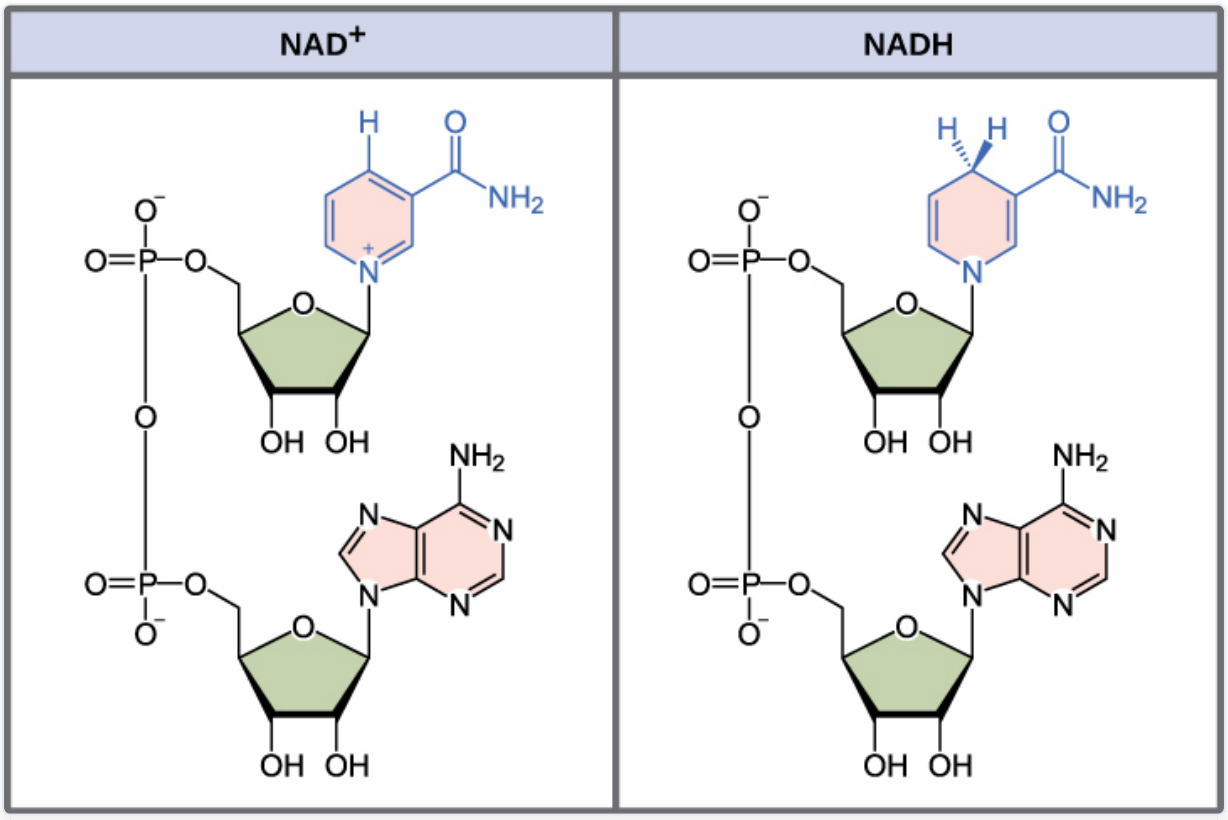
Electrons and Energy

The removal of an electron from a molecule (oxidizing it), results in a decrease in **potential energy** in the oxidized compound. However, the electron (sometimes as part of a hydrogen atom) does not remain unbonded in the cytoplasm of a cell. Rather, the electron is shifted to a second compound, reducing the second compound. *The shift of an electron from one compound to another removes some potential energy from the first compound (the oxidized compound) and increases the potential energy of the second compound (the reduced compound)*. The transfer of electrons between molecules is important because most of the energy stored in atoms and used to fuel cell functions is in the form of high-energy electrons. The transfer of energy in the form of high-energy electrons allows the cell to transfer and use energy in an incremental fashion—in small packages rather than in a single, destructive burst. This chapter focuses on the extraction of energy from food; you will see that as you track the path of the transfers, you are tracking the path of electrons moving through metabolic pathways.

Electron Carriers

In living systems, a small class of compounds functions as electron carriers: they bind and carry high-energy electrons between compounds in biochemical pathways. These compounds can be easily reduced (that is, they accept electrons) or oxidized (they lose electrons). NAD^+ is the oxidized form of the molecule NAD; NADH is the reduced form of the molecule after it has accepted two electrons and a proton. Note that if a compound has an “H” on it, it is generally reduced (e.g., NADH is the reduced form of NAD). When electrons are added to a compound, *it is reduced*. A compound that reduces another is called a reducing agent. In the above equation,

Similarly, flavin adenine dinucleotide (FAD^+) is also an electron carrier. Its reduced form is $FADH_2$. Both NAD^+ and FAD^+ are extensively used in energy extraction from sugars.



This illustration shows the molecular structure of N A D superscript plus sign baseline and N A D H. Both compounds are composed of an adenine nucleotide and a nicotinamide nucleotide, which bond together to form a dinuc...

Figure 7.3 The oxidized form of the electron carrier (NAD^+) is shown on the left, and the reduced form (NADH) is shown on the right. The nitrogenous base in NADH has one more hydrogen ion and two more electrons than in NAD^+.

ATP in Living Systems

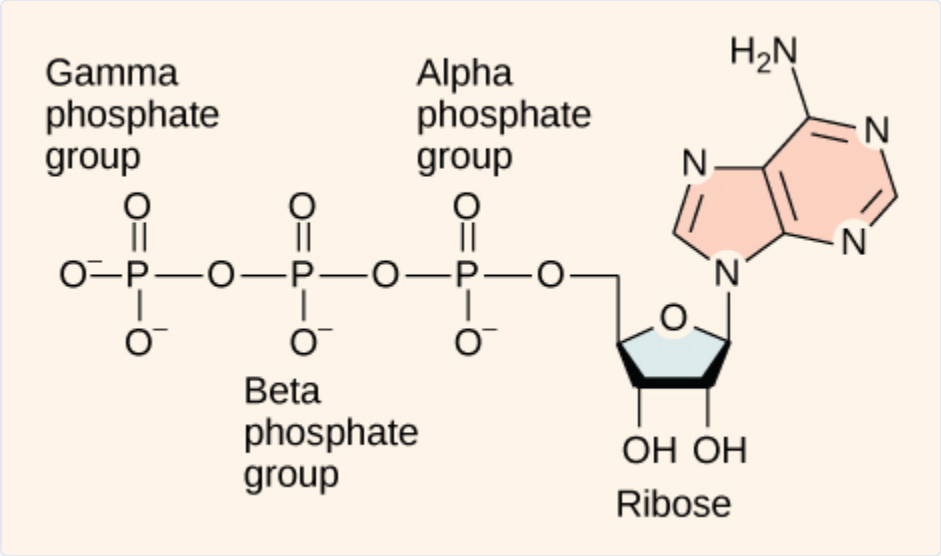
A living cell cannot store significant amounts of free energy. Excess free energy would result in an increase of heat in the cell, which would result in excessive thermal motion that could damage and then destroy the cell. Rather, a cell must be able to handle that energy in a way that enables the cell to store energy safely and release it for use only as needed. Living cells accomplish this by using the compound adenosine triphosphate (ATP). ATP is often called the "energy currency" of the cell, and, like currency, this versatile compound can be used to fill any energy need of the cell. How? It functions similarly to a rechargeable battery.

When ATP is broken down, by the removal of its terminal phosphate group, energy is released. The energy is used to do work by the cell, usually when the released phosphate binds to another molecule, thereby activating it. Recall the active transport work of the sodium-potassium pump in cell membranes. ATP alters the structure of the integral protein that functions as the pump, changing its affinity for sodium and potassium. In this way, the cell performs work, pumping ions against their electrochemical gradients.

ATP Structure and Function

At the heart of ATP is a molecule of adenosine monophosphate (AMP), which is composed of an adenine molecule bonded to a ribose molecule and to a single phosphate group (Figure 7.4). Ribose is a five-carbon sugar found in RNA, and AMP is one of the nucleotides in RNA. The addition of a second phosphate group to this core molecule

results in the formation of adenosine diphosphate (ADP); the addition of a third phosphate group forms adenosine triphosphate (ATP).



This illustration shows the molecular structure of A T P. This molecule is an adenine nucleotide with a string of three phosphate groups attached to it. The phosphate groups are named alpha, beta, and gamma in order of i...

Figure 7.4 ATP (adenosine triphosphate) has three phosphate groups that can be removed by hydrolysis (addition of H_2O) to form ADP (adenosine diphosphate) or AMP (adenosine monophosphate). The negative charges on the phosphate group naturally repel each other, requiring energy to bond them together and releasing energy when these bonds are broken.

The addition of a phosphate group to a molecule requires energy. Phosphate groups are negatively charged and thus repel one another when they are arranged in series, as they are in ADP and ATP. This repulsion makes the ADP and ATP molecules inherently unstable. The release of one phosphate group from ATP, a process called dephosphorylation, releases energy.

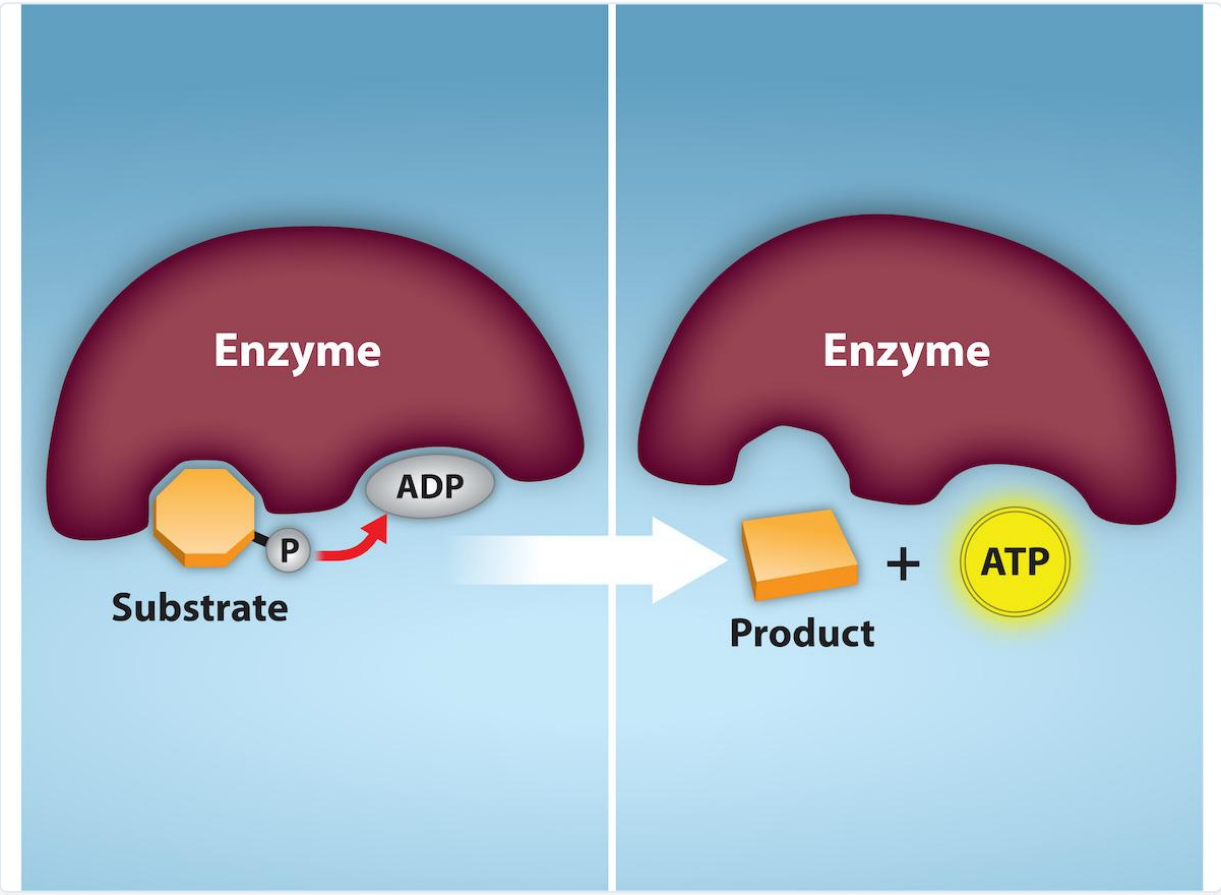
Energy from ATP

Hydrolysis is the process of breaking complex macromolecules apart. During hydrolysis, water is split, or lysed, and the resulting hydrogen atom (H^+) and a hydroxyl group (OH^-), or *hydroxide,* are added to the larger molecule. The hydrolysis of ATP produces ADP, together with an inorganic phosphate ion (P_i), and the release of free energy. To carry out life processes, ATP is continuously broken down into ADP, and like a rechargeable battery, ADP is continuously regenerated into ATP by the reattachment of a third phosphate group. Water, which was broken down into its hydrogen atom and hydroxyl group (hydroxide) during ATP hydrolysis, is regenerated when a third phosphate is added to the ADP molecule, reforming ATP.

Obviously, energy must be infused into the system to regenerate ATP. Where does this energy come from? In nearly every living thing on Earth, the energy comes from the metabolism of glucose, fructose, or galactose, all isomers with the chemical formula $C_6H_{12}O_6$ but different molecular configurations. In this way, ATP is a direct link between the limited set of exergonic pathways of glucose catabolism and the multitude of endergonic pathways that power living cells.

Substrate-Level Phosphorylation

ATP is generated through two mechanisms during the breakdown of glucose. A few ATP molecules are generated (that is, regenerated from ADP) as a direct result of the chemical reactions that occur in the catabolic pathways. A phosphate group is removed from an intermediate reactant in the pathway, and the free energy of the reaction is used to add the third phosphate to an available ADP molecule, producing ATP (Figure 7.5). This very direct method of phosphorylation is called **substrate-level phosphorylation**.

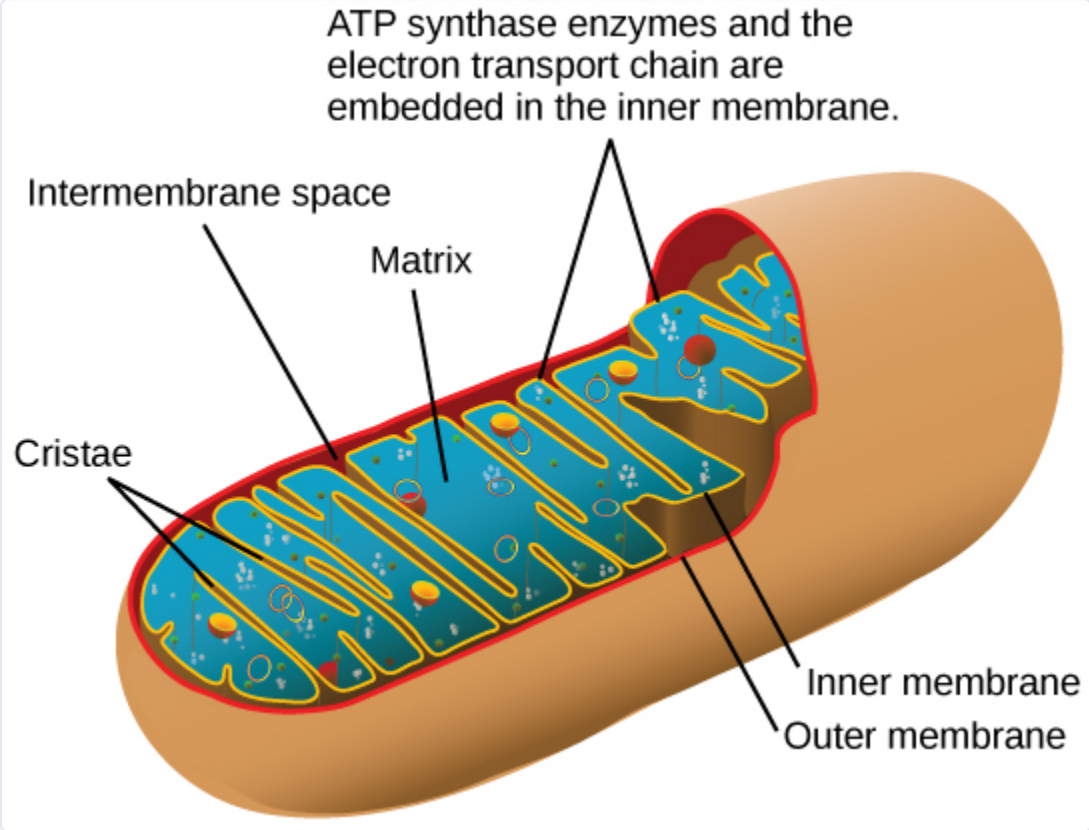


This illustration shows a substrate-level phosphorylation reaction in which the gamma phosphate of A T P is attached to a protein. The substrate, phosphate group, and A D P bind to an enzyme. And the result is the produc...

Figure 7.5 In phosphorylation reactions, the gamma (third) phosphate of ATP is attached to a protein. In substrate-level phosphorylation, a phosphate group that is covalently attached to another molecule is transferred to ADP to form ATP. Credit: Rao, A., Ryan, K. and Fletcher, S. Department of Biology, Texas A&M University.

Oxidative Phosphorylation

Most of the ATP generated during cellular respiration, however, is derived from a much more complex process, chemiosmosis, which takes place in mitochondria (Figure 7.6) within a eukaryotic cell or the plasma membrane of a prokaryotic cell. Chemiosmosis, a process of ATP production in cellular metabolism, is used to generate 90 percent of the ATP made during cellular respiration and is also the method used in the light reactions of photosynthesis to harness the energy of sunlight. The production of ATP using the process of chemiosmosis is called **oxidative phosphorylation** because of the involvement of oxygen in the process.



This illustration shows the structure of a mitochondrion, which has an outer membrane and an inner membrane. The inner membrane has many folds, called cristae. The space between the outer membrane and the inner membrane ...

Figure 7.6 In eukaryotes, oxidative phosphorylation takes place in mitochondria. In prokaryotes, this process takes place in the plasma membrane. (Credit: modification of work by Mariana Ruiz Villareal)

Section 9.2: Glycolysis (The Cytosolic Splitting of Glucose)

Energy investment phase, fructose-1,6-bisphosphate cleavage, energy payoff phase, and net stoichiometric products (2 Pyruvate, 2 Net ATP, 2 NADH).

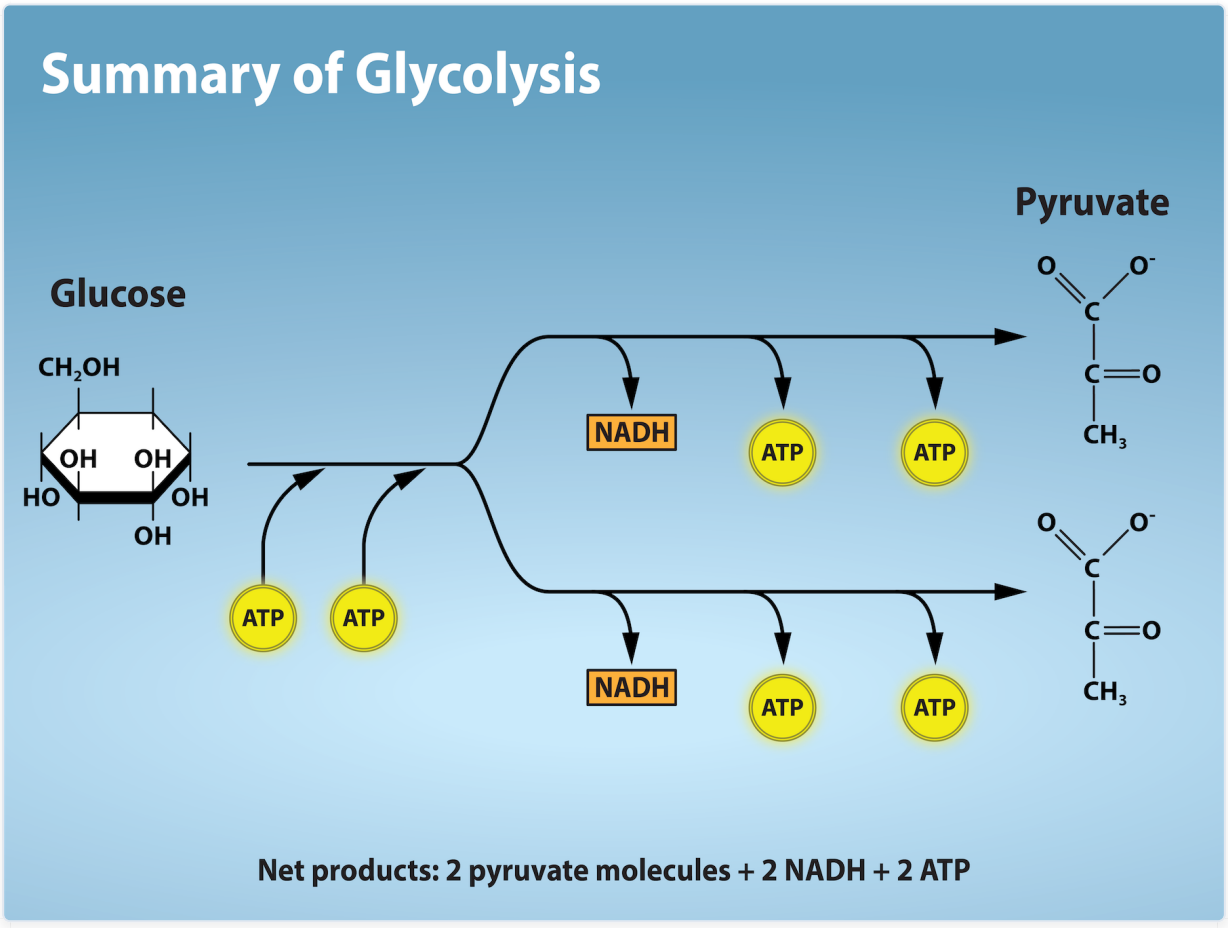
Learning Objectives

By the end of this section, you will be able to do the following:

- Describe the overall result in terms of molecules produced during the chemical breakdown of glucose by glycolysis
- Compare the output of glycolysis in terms of ATP molecules and NADH molecules produced

As you have read, nearly all of the energy used by living cells comes to them in the bonds of the sugar glucose. **Glycolysis** is the first step in the breakdown of glucose to extract energy for cellular metabolism. In fact, nearly all living organisms carry out glycolysis as part of their metabolism. Glycolysis takes place in the cytoplasm of both prokaryotic and eukaryotic cells.

Glycolysis begins with the six-carbon ring-shaped structure of a single glucose molecule and ends with two molecules of a three-carbon sugar called **pyruvate**. Glycolysis consists of two distinct phases. The first part of the glycolysis pathway traps the glucose molecule in the cell and **uses energy** to modify it so that the six-carbon sugar molecule can be split evenly into the two three-carbon molecules. The second part of glycolysis **extracts energy** from the molecules and stores it in the form of ATP and NADH—remember: this is the reduced form of NAD.



The illustration shows a simplified process of glucose moving through the stages of glycolysis. First two A T P are added, then the glucose is split into two branches, with N A D H and two A T P being released. The net ...

Figure 7.7 Glycolysis begins with an energy investment phase which requires 2 ATP to phosphorylate the starting glucose molecule. The 6-carbon intermediate is then split into 2, 3-carbon sugar molecules. In the energy recovery phase, each 3-carbon sugar is then oxidized to pyruvate with the energy transferred to form NADH and 2 ATP. Credit: Rao, A. and Ryan, K. Department of Biology, Texas A&M University

Outcomes of Glycolysis

Glycolysis begins with glucose and produces two pyruvate molecules, four new ATP molecules, and two molecules of NADH. (Note: two ATP molecules are used in the first half of the pathway to prepare the six-carbon ring for cleavage, so the cell has a ***net gain of two ATP molecules*** and two NADH molecules for its use). If the cell cannot catabolize the pyruvate molecules further, it will harvest only two ATP molecules from one molecule of glucose.

Section 9.3: Pyruvate Oxidation (The Mitochondrial Entryway)

Mitochondrial transport carrier, decarboxylation (-CO_2), NAD^+ reduction to NADH, and Coenzyme A attachment forming Acetyl-CoA.

Learning Objectives

By the end of this section, you will be able to do the following:

- Describe how pyruvate, the product of glycolysis, is prepared for entry into the citric acid cycle

If oxygen is available, aerobic respiration will go forward. In eukaryotic cells, the pyruvate molecules produced at the end of glycolysis are transported into the mitochondria, which are the sites of cellular respiration. There, pyruvate is transformed into an acetyl group that will be picked up and activated by a carrier compound called coenzyme A (CoA). The resulting compound is called **acetyl CoA.** Acetyl CoA can be used in a variety of ways by the cell, but its major function is to deliver the acetyl group derived from pyruvate to the next stage of the pathway in glucose catabolism: The Citric Acid Cycle (CAC).

Breakdown of Pyruvate

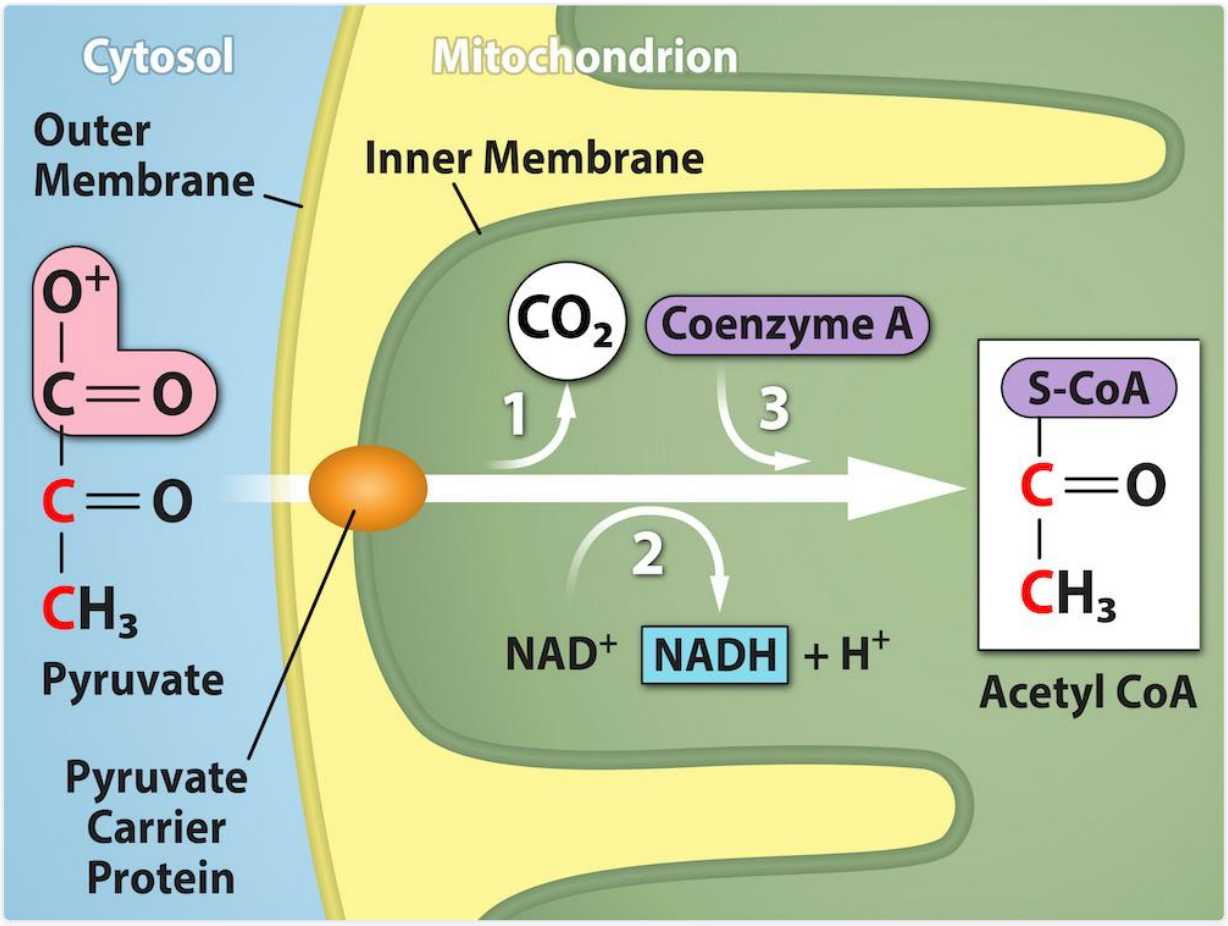
In order for pyruvate, the product of glycolysis, to enter the next pathway, it must undergo several changes. The conversion is a three-step process (Figure 7.10).

Step 1. A carboxyl group is removed from pyruvate, releasing a molecule of **carbon dioxide**. This reaction creates a two-carbon molecule (called hydroxyethyl).

Step 2. The two carbon molecule is oxidized to an acetyl group, and the electrons are picked up by NAD^+, forming **NADH**. The high-energy electrons from NADH will be used later to generate ATP. Th

Step 3. The enzyme-bound acetyl group is transferred to CoA, producing a molecule of **acetyl CoA**.

These three step happens twice, once for each pyruvate molecule. Remember, there were two pyruvate molecules produced at the end of glycolsis. It should noted that these three steps (happening twice) results in the removal of two carbons from original glucose molecule.



This illustration shows the conversion of pyruvate into acetyl upper case C lower case o upper case A. The Pyruvate begins outside the mitochondrion, in the cytosol. When the pyruvate enters the mitochondrion through the...

Figure 7.10 Upon entering the mitochondrial matrix, a multienzyme complex converts pyruvate into acetyl CoA. In the process, carbon dioxide is released, and one molecule of NADH is formed. Credit: Rao, A., Ryan, K. and Tag, A. Department of Biology, Texas A&M University.

Note that during the second stage of glucose metabolism, whenever a carbon atom is removed, it is bound to two oxygen atoms, producing **carbon dioxide**, one of the major end products of cellular respiration.

The two newly produced Acetyl CoA molecules will now progress to the next step in Cellular Respiration: The Citric Acid Cycle (CAC).

Section 9.4: The Citric Acid Cycle (Krebs Cycle / TCA Cycle)

8-step enzymatic cyclic oxidation of Acetyl-CoA, oxaloacetate condensation, GTP/ATP substrate-level phosphorylation, and NADH/$FADH_2$ yields.

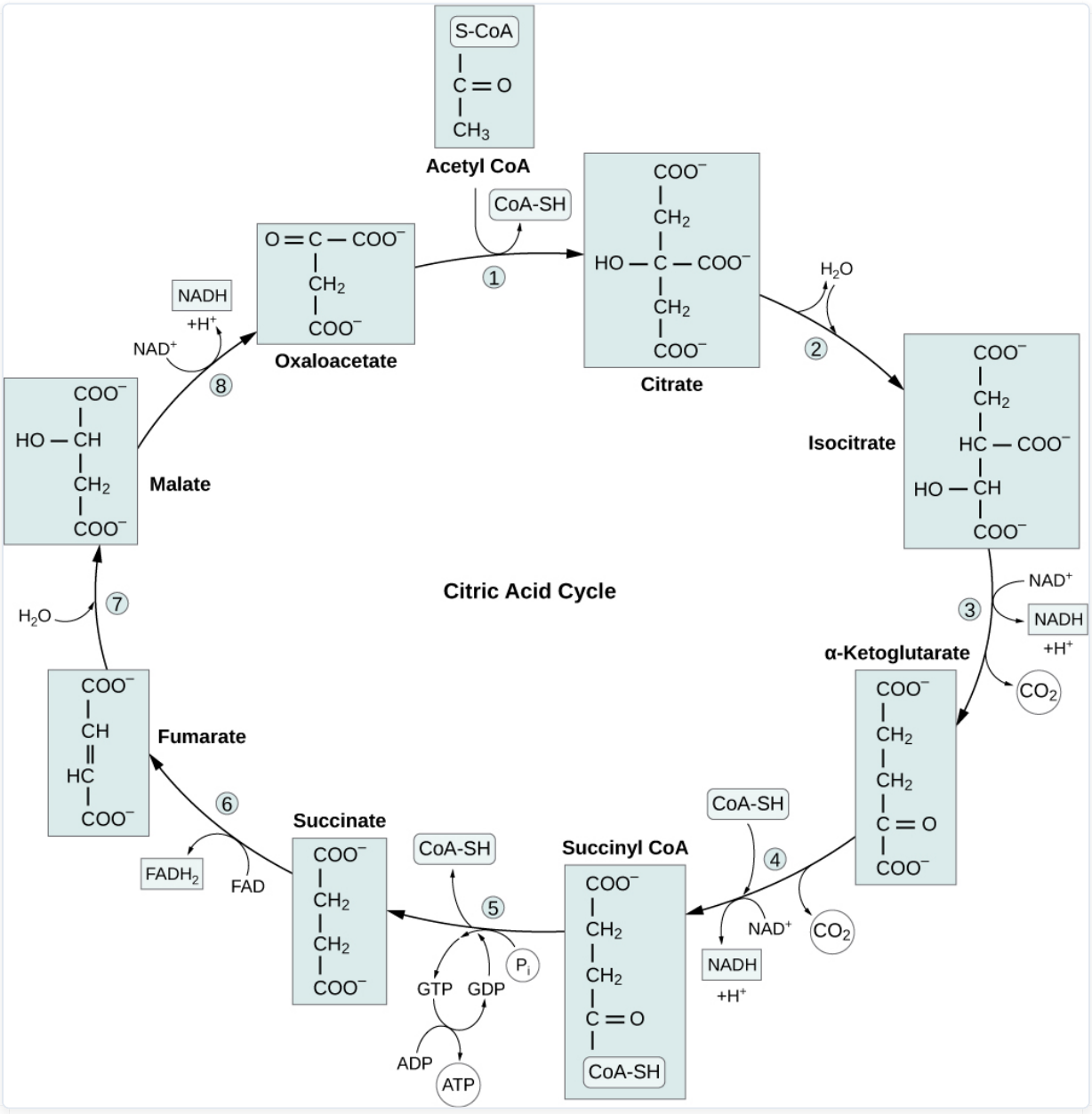
Learning Objectives

By the end of this section, you will be able to do the following:

- Explain how a circular pathway, such as the citric acid cycle, fundamentally differs from a linear biochemical pathway, such as glycolysis

Citric Acid Cycle

Like the conversion of pyruvate to acetyl CoA, the citric acid cycle takes place in the matrix of mitochondria. Unlike glycolysis, the citric acid cycle is a closed loop: the last part of the pathway regenerates the compound used in the first step. The steps of the cycle are a series of redox reactions that produce two carbon dioxide molecules, one ATP, and the reduced carriers 3 NADH and one $FADH_2$ (Figure 7.11). Note that the citric acid cycle produces very little ATP directly. Both Acetyl CoA molecules proceed through the Citric Acid Cycle which means the total outcome is four carbon dioxide molecules, 2 ATP, 6 NADH, and 2 $FADH_2$.



This illustration shows the eight steps of the citric acid cycle. In the first step, the acetyl group from acetyl uppercase C lower case o upper case A is transferred to a four-carbon oxaloacetate molecule to form a six-...

Figure 7.11 In the citric acid cycle, the acetyl group from acetyl CoA is attached to a four-carbon oxaloacetate molecule to form a six-carbon citrate molecule. Through a series of steps, citrate is oxidized, releasing two carbon dioxide molecules for each acetyl group fed into the cycle. In the process, three NAD^+ molecules are reduced to NADH, one FAD molecule is reduced to $FADH_2$, and one ATP or GTP (depending on the cell type) is produced (by substrate-level phosphorylation). Because the final product of the citric acid cycle is also the first reactant, the cycle runs continuously in the presence of sufficient reactants. Credit: Rao, A., Ryan, K., Tag, A., and Fletcher, S. Department of Biology, Texas A&M University.

Section 9.5: Oxidative Phosphorylation & Chemiosmosis

Electron transport chain complexes I–IV, Coenzyme Q, Cytochrome c, terminal O_2 reduction to H_2O, and ATP synthase rotary chemiosmosis.

Learning Objectives

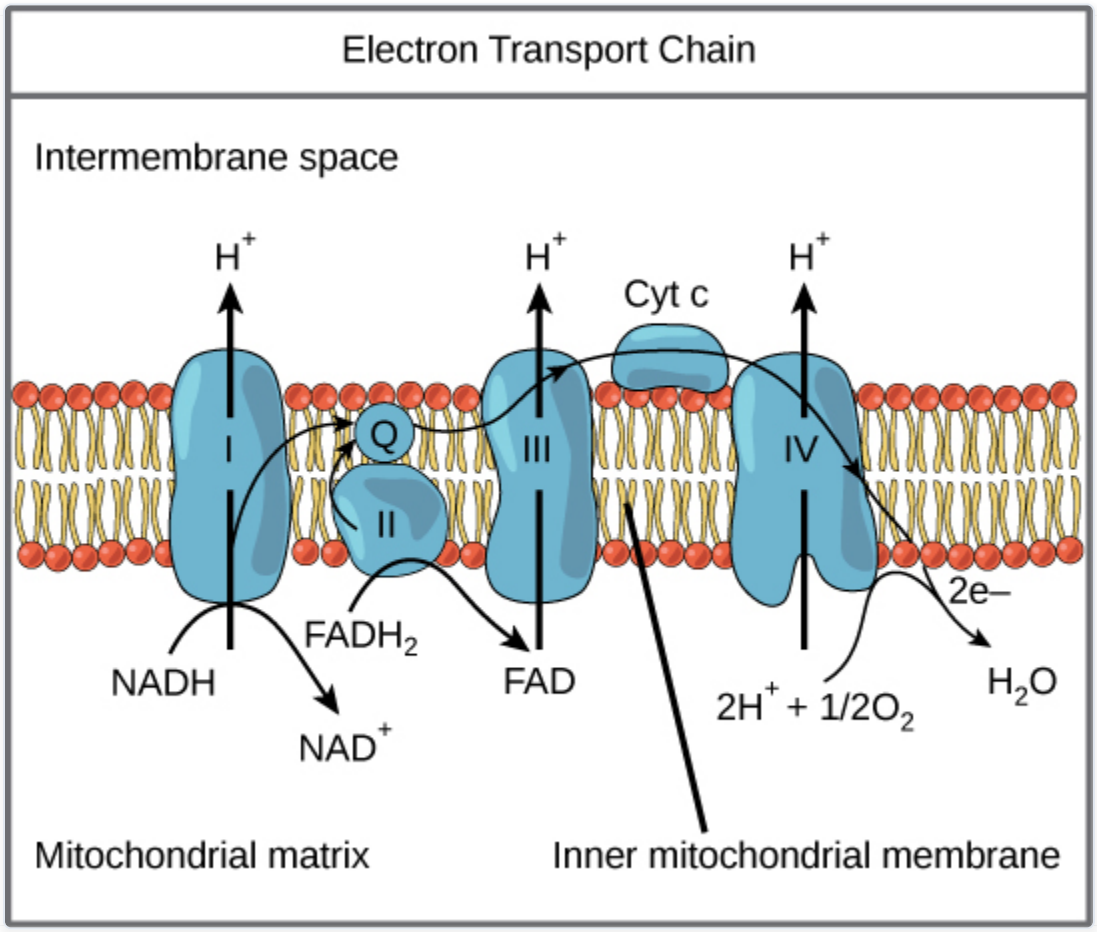
By the end of this section, you will be able to do the following:

- Describe how electrons move through the electron transport chain and explain what happens to their energy levels during this process
- Explain how a proton (H^+) gradient is established and maintained by the electron transport chain

You have just read about two pathways in glucose catabolism that generate ATP —glycolysis and the citric acid cycle. Most of the ATP generated during the aerobic catabolism of glucose, however, is not generated directly from these pathways. Instead, it is derived from a process that begins by moving electrons through a series of electron carriers. This process causes hydrogen ions to accumulate within the mitochondrial intermembrane space. Therefore, a hydrogen ion concentration gradient forms in mitochondrial intermembrane space. Eventually, those hydrogen ions will diffuse out of the intermembrane space into the mitochondrial matrix by passing through **ATP Synthase**. The current of hydrogen ions through ATP Synthase powers the catalytic action of ATP synthase, which phosphorylates ADP, producing ATP.

Electron Transport Chain

The electron transport chain (ETC) (Figure 7.12) is the last component of aerobic respiration and is the only part of glucose metabolism that uses atmospheric oxygen. In animals, oxygen enters the body through the respiratory systems, this is why animals must breath oxygen. The electron transport chain is a series of redox reactions that resembles a relay race or bucket brigade in that electrons are passed rapidly from one component to the next, to the endpoint of the chain where the electrons reduce oxygen to bond to hydrogen ions, which produces water. There are four complexes composed of proteins, labeled I through IV in Figure 7.12 , and the aggregation of these four complexes, together with associated mobile, electron shuttles, is called the electron transport chain. The electron transport chain is present with multiple copies in the mitochondrial inner membrane of eukaryotes.



This illustration shows the electron transport chain embedded in the inner mitochondrial membrane. The electron transport chain consists of four electron complexes. Complex I oxidizes N A D H to N A D superscript plus si...

Figure 7.12 The electron transport chain is a series of electron transporters embedded in the inner mitochondrial membrane that shuttles electrons from NADH and $FADH_2$ to molecular oxygen. In the process, protons are pumped from the mitochondrial matrix to the intermembrane space, and oxygen is reduced to form water.

Complex I

First, NADH brings two electrons to Complex I. This reduces Complex I and oxidizes NADH back to NAD+. Complex I can pump four hydrogen ions across the membrane from the matrix into the intermembrane space, which begins to establish a hydrogen ion gradient in the intermembrane space.

Q and Complex II

Complex II receives $FADH_2$ ($FADH_2$ does not go to Complex I). Ubiquinon (Q) is an electron shuttle that connects Complex I to Complex III, and Complex II with Complex III. Therefore, the electrons from NADH are transferred from Complex I to Complex III, and the electrons from $FADH_2$ are transferred from Complex II to Complex III. Once it is reduced (by accepting the electrons from Complex I and Complex II) ubiquinone (Q) delivers its electrons to the next complex in the electron transport chain, Complex III. Q receives the electrons derived from NADH from complex I, and the electrons derived from $FADH_2$ from complex II.

Complex III

Another electron shuttle called Cytochrome C (Cyt C) transfers electrons from Complex III to Complex IV. The oxidation of Complex III produces energy that is used to pump hydrogen ions from the matrix into the intermembrane space, continuing to build a hydrogen ion concentration gradient.

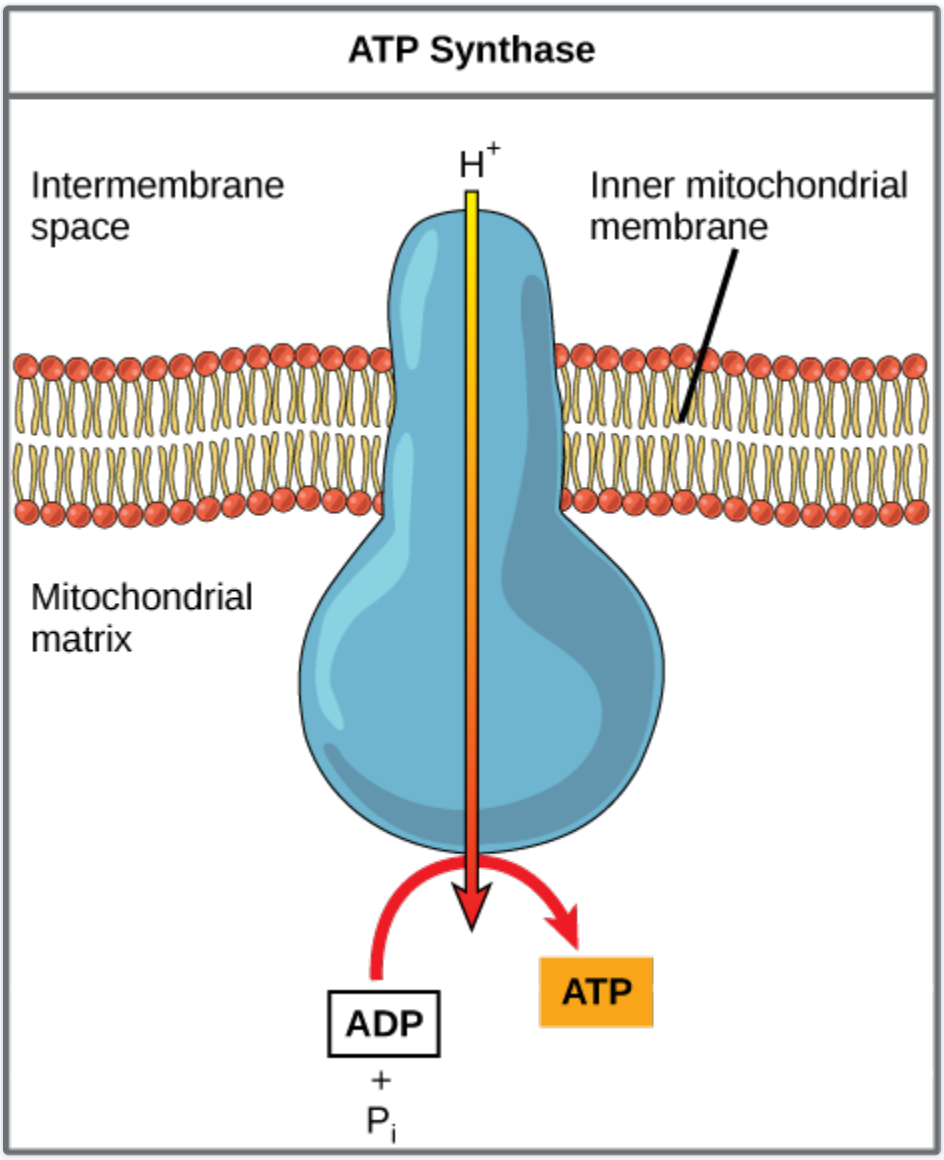
Complex IV

Complex IV receives electrons from the shuttle Cytochrome C, reducing Complex IV and oxidizing Cytochrome C. Complex IV then transfers its electrons to oxygen, reducing oxygen and oxidizing Complex IV. The reduced oxygen then picks up two hydrogen ions from the matrix to make water (H_2O). **Oxygen is the final electron acceptor in the ETC**. The removal of the hydrogen ions from the system contributes to the ion gradient that forms the foundation for the process of chemiosmosis.

Chemiosmosis

In chemiosmosis, the free energy from the series of redox reactions of the ETC was used to pump hydrogen ions (protons) across the mitochondrial membrane, into the intermembrane space. The uneven distribution of H^+ ions across the membrane establishes a concentration gradient.

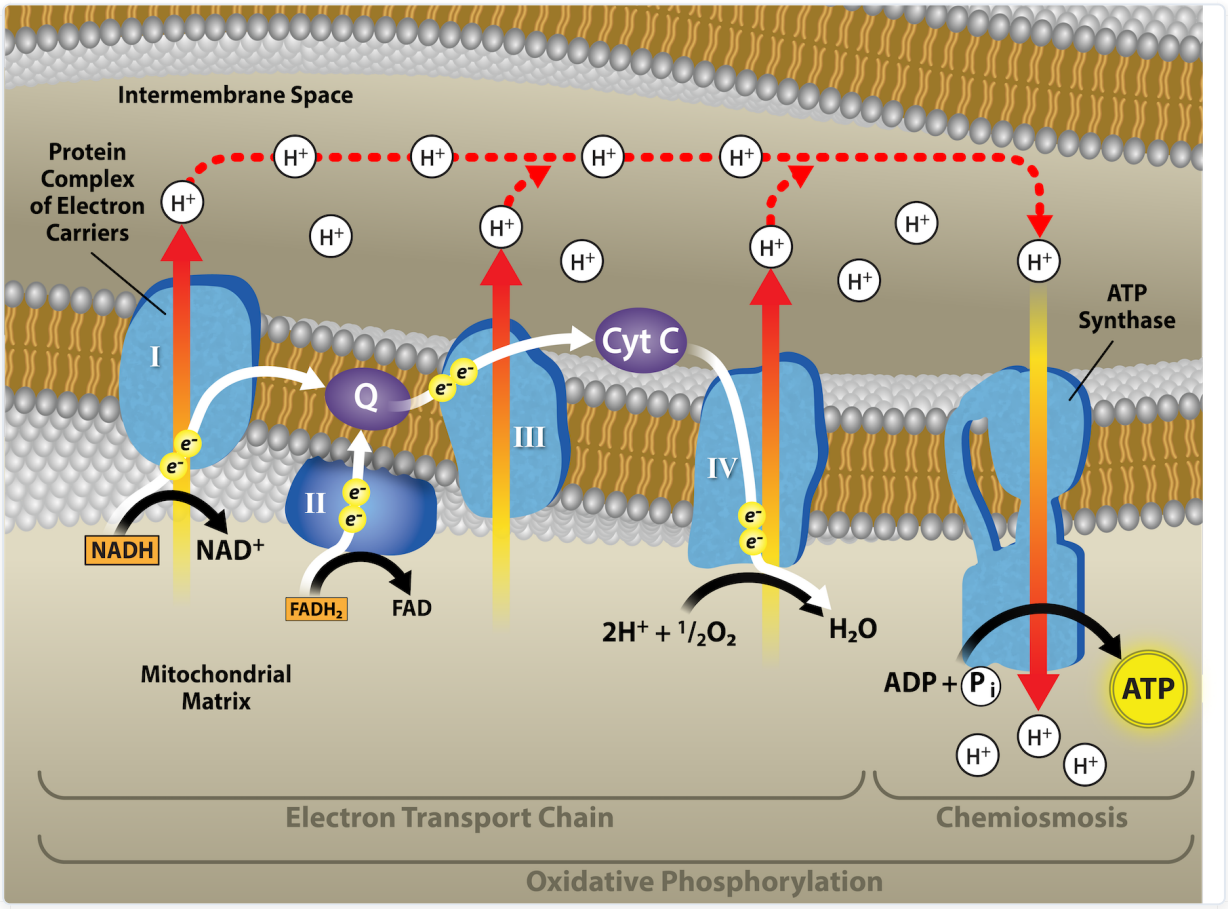
If the membrane were continuously open to simple diffusion by the hydrogen ions, the ions would tend to diffuse back across into the matrix, driven by the concentrations producing their electrochemical gradient. Recall that ions cannot diffuse through the nonpolar regions of phospholipid membranes without the aid of membrane proteins. Thus, hydrogen ions in the intermembrane space require a transmembrane integral transport protein to move into the matrix. This transport protein is called **ATP synthase** (Figure 7.13). This complex protein acts as a tiny generator, turned by the force of the hydrogen ions diffusing through it, down their concentration gradient. The turning of parts of this molecular machine facilitates the addition of a phosphate to ADP, forming ATP, *using the potential energy of the hydrogen ion gradient*. This process is called **chemiosmosis**.



This illustration shows an A T P synthase enzyme embedded in the inner mitochondrial membrane. A T P synthase allows protons to move from an area of high concentration in the intermembrane space to an area of low concent...

Figure 7.13 ATP synthase is a complex, molecular machine that uses a proton (H^+) gradient to form ATP from ADP and inorganic phosphate (Pi). (Credit: modification of work by Klaus Hoffmeier)

Chemiosmosis (Figure 7.14) is used to generate 90 percent of the ATP made during aerobic glucose catabolism. ATP produced during chemiosmosis is oxidative phosphorylation. The overall result of these reactions is the production of ATP from the energy of the electrons removed from hydrogen atoms. These atoms were originally part of the original glucose molecule. The number of ATP generated by cellular respiration varies depending on the organism and the type of cell undergoing cellular respiration. On average, roughly 26 - 28 molecules of ATP are produced during Oxidative Phosphorylation.



This illustration shows the electron transport chain, the A T P synthase enzyme embedded in the inner mitochondrial membrane, and chemiosmosis occurring in the mitochondrial matrix. The electron transport chain oxidizes...

Figure 7.14 In oxidative phosphorylation, the pH gradient formed by the electron transport chain is used by ATP synthase to form ATP. Credit: Rao, A., Ryan, K., Fletcher, S. and Tag, A. Department of Biology, Texas A&M University.

Section 9.6: Instructor Lecture Slide Review (Chapter 9_students Part 1 & 2)

Official 49-slide lecture deck synthesis, catabolic redox pathways, and aerobic energy accounting.

Lecture Part 1 • Cellular Respiration & Catabolic Redox Mechanics

Cellular respiration is a series of catabolic redox reactions that harvest chemical potential energy stored in organic bonds to synthesize ATP. Oxidation is the loss of electrons (LEO); reduction is the gain of electrons (GER). Glucose is oxidized to CO_2 while O_2 is reduced to H_2O. NAD^+ and FAD serve as coenzymes that accept high-energy electrons (reducing to NADH and $FADH_2$) for delivery to the inner mitochondrial membrane.

Lecture Part 1 • Glycolysis, Pyruvate Oxidation & The Krebs Cycle

Glycolysis occurs in the cytosol without requiring O_2. Investment of 2 ATP produces 4 ATP (via substrate-level phosphorylation) and 2 NADH, yielding 2 net ATP and 2 Pyruvate. In aerobic cells, pyruvate is actively transported into the mitochondrial matrix. Pyruvate dehydrogenase converts each 3C pyruvate into a 2C Acetyl-CoA, releasing CO_2 and producing 1 NADH (2 NADH per glucose). Acetyl-CoA enters the Citric Acid Cycle, condensing with 4C oxaloacetate to form 6C citrate. Per glucose (2 turns), the CAC yields 4 CO_2, 6 NADH, 2 $FADH_2$, and 2 ATP.

Lecture Part 2 • Oxidative Phosphorylation & Cellular Poisons

NADH transfers electrons to Complex I, whereas $FADH_2$ enters downstream at Complex II. Electrons travel through ubiquinone (Q), Complex III, cytochrome c, and Complex IV to reach the terminal electron acceptor O_2, forming H_2O. As electrons flow, Complexes I, III, and IV pump protons from the matrix into the intermembrane space. Protons diffuse back through ATP synthase (chemiosmosis), generating ~26–28 ATP. Uncouplers (like DNP) dissipate the proton gradient, producing heat without ATP. Inhibitors like rotenone block Complex I; oligomycin blocks ATP synthase.

Lecture Part 2 • Fermentation Pathways

Without oxygen, electron transport halts and NADH accumulates. Fermentation allows cells to regenerate NAD^+ from NADH in the cytosol to keep glycolysis running. In *lactic acid fermentation*, pyruvate is reduced directly to lactate by NADH (muscles, yogurt). In *alcohol fermentation*, pyruvate is decarboxylated to acetaldehyde, which is reduced to ethanol, releasing CO_2 (yeast brewing/baking).

Section 9.7: Official Course Practice Questions & Solutions

All 27 practice examination problems from Chapter 9 Practice Questions.docx with verified answers and explanations.

1. Where in the cell does glycolysis take place?

A) Nucleus • B) Intermembrane space • C) Mitochondria • **D) Cytosol/Cytoplasm (Correct)** • E) Golgi body

2. Which of the following products is not created during glycolysis?

A) ATP • B) NADH • **C) Oxygen (Correct - O_2 is not produced)** • D) Pyruvate

3. What is the net production of ATP from one glucose molecule in glycolysis?

A) 4 • **B) 2 (Correct: 4 produced - 2 invested = 2 net)** • C) 28 • D) 32

4. What are the net products of glycolysis?

A) 2 pyruvate, 1 $FADH_2$, 2 ATP • B) 2 pyruvate, 2 NADH, 4 ATP • **C) 2 pyruvate, 2 NADH, 2 ATP (Correct)** • D) 4 pyruvate, 2 $FADH_2$, 4 ATP

5. A solution in a beaker contains all enzymes for glycolysis. Glucose is mixed in, O_2 is present, but NO ATP is present. Does glycolysis occur?

A) Yes • **B) No (Correct: Glycolysis requires initial investment of 2 ATP to phosphorylate glucose)**

6. A metabolic pathway that releases energy by breaking down complex molecules to simpler compounds is known as:

A) Anabolic • B) Catabolic • C) Photosynthesis • D) Cellular Respiration • **E) B & D (Correct: Cellular respiration is catabolic)**

7. What is the purpose of electron carriers NADH & $FADH_2$?

A) Transport electrons out of mitochondria • **B) To deliver electrons to the electron transport chain (Correct)** • C) Create water • D) Neutralize charge

8. What is the final electron acceptor in the electron transport chain?

A) NAD^+ • **B) Oxygen (O_2) (Correct)** • C) Water • D) ATP Synthase

9. Which of the following describes oxidative phosphorylation?

A) Formation of ATP using energy derived from redox reactions of the ETC and chemiosmosis (Correct) • B) Direct transfer of phosphate from substrate to ADP • C) Formation of glucose • D) Hydrolysis of water

10. During electron transport, protons are pumped from where to where?

A) Intermembrane space to matrix • B) Cytosol to matrix • **C) Mitochondrial matrix into the intermembrane space (Correct)** • D) Matrix to cytosol

11. What directly drives the synthesis of ATP by ATP synthase?

A) Direct transfer of phosphate from NADH • **B) The flow of H^+ ions down their electrochemical gradient from the intermembrane space into the matrix (Correct)** • C) O_2 binding • D) Glucose hydrolysis

12. How many net ATP molecules are typically produced from one molecule of glucose in aerobic cellular respiration?

A) 2 • **B) 30–32 (Correct)** • C) 4 • D) 38–40

13. In the absence of oxygen, yeast cells undergo alcohol fermentation. What are the products?

A) Lactic acid and ATP • B) CO_2 and H_2O • **C) Ethanol, CO_2, and NAD^+ (Correct)** • D) Pyruvate and NADH

14. What is the primary functional role of fermentation in muscle cells?

A) To produce maximum ATP • B) To produce O_2 • **C) To regenerate NAD^+ so that glycolysis can continue producing ATP (Correct)** • D) To store glycogen

15. How many molecules of CO_2 are produced from the complete cellular respiration of ONE glucose molecule?

A) 2 • B) 4 • **C) 6 (Correct: 2 in pyruvate oxidation + 4 in Citric Acid Cycle)** • D) 12

16. Where does pyruvate oxidation take place in eukaryotic cells?

A) Cytosol • B) Intermembrane space • C) Outer membrane • **D) Mitochondrial Matrix (Correct)**

17. Where does the Citric Acid Cycle take place?

A) Cytosol • B) Cristae • C) Intermembrane space • **D) Mitochondrial Matrix (Correct)**

18. Where are the electron transport chain protein complexes located?

A) Inner mitochondrial membrane (Cristae) (Correct) • B) Outer membrane • C) Matrix • D) Intermembrane space

19. How many pyruvates are produced from the breakdown of one glucose molecule?

A) 1 • **B) 2 (Correct)** • C) 3 • D) 4

20. What is produced from ONE turn of the Citric Acid Cycle per Acetyl-CoA?

A) 6 NADH, 2 $FADH_2$, 2 ATP • **B) 3 NADH, 1 $FADH_2$, 1 ATP, 2 CO_2 (Correct)** • C) 2 net ATP, 2 NADH • D) 2 NADH, 2 CO_2

21. Which phase(s) of cellular respiration produce ATP via substrate-level phosphorylation?

Answer: Glycolysis (in the cytosol) and the Citric Acid Cycle (in the mitochondrial matrix).

22. Which phase(s) produce ATP via oxidative phosphorylation?

Answer: Oxidative Phosphorylation (the combination of the Electron Transport Chain and Chemiosmosis across the inner mitochondrial membrane).

23. What is the fundamental difference between substrate-level phosphorylation and oxidative phosphorylation?

Answer: Substrate-level phosphorylation uses a soluble kinase enzyme to directly transfer a phosphate group from a phosphorylated substrate to ADP. Oxidative phosphorylation uses energy from electron redox transfers to generate a proton gradient that powers ATP synthase to add inorganic phosphate (P_i) to ADP.

Appendix: Course Active Learning Practice Sets & Solutions

Original in-class problem sets, stoichiometric pathway accounting, and cellular poisons diagnostics from Active Learning #3 (Slides 4–7).

Problem 1: Glycolysis & Pyruvate Oxidation Accounting (Active Learning #3, Slide 4)

Reaction Stage	Direct Product Yield	Subcellular Destination & Metabolic Fate
Glycolysis (in Cytosol)	**2 Pyruvate**	Transported through mitochondrial carrier into matrix for Stage 2 (Pyruvate Oxidation).
	4 ATP (2 Net ATP)	Generated via substrate-level phosphorylation; immediately used for cellular work.
	2 NADH	Shuttled into mitochondria to donate high-energy electrons to the ETC.
Pyruvate Oxidation (per Pyruvate; x2 per Glucose)	**1 Acetyl-CoA** (2 per glucose)	Stays in mitochondrial matrix to enter the Citric Acid Cycle (Stage 3).
	1 NADH (2 per glucose)	Stays in mitochondrial matrix; donates electrons directly to Complex I of the ETC.
	1 CO_2 (2 per glucose)	Diffuses out of the mitochondrion, into the bloodstream, and is exhaled.

Problem 2: The Citric Acid Cycle Accounting (Active Learning #3, Slide 5)

Product Molecule	Yield per Acetyl-CoA	Yield per Glucose & Cellular Destination
NADH	3 NADH	**6 NADH per glucose**; delivers high-energy electrons to ETC Complex I in matrix.
$FADH_2$	1 $FADH_2$	**2 $FADH_2$ per glucose**; delivers electrons to ETC Complex II.
ATP (via GTP)	1 ATP	**2 ATP per glucose**; generated via substrate-level phosphorylation for cellular work.
CO_2	2 CO_2	**4 CO_2 per glucose**; fully oxidized waste product; diffused and exhaled.

Problem 3: Oxidative Phosphorylation & Chemiosmosis (Active Learning #3, Slide 6)

Mechanistic Question	Biochemical Answer & Instructor Rationale
H^+ Transport Direction	Protons (H^+) are actively transported from the **Mitochondrial Matrix** into the **Intermembrane Space**.
Transport Classification	**Active Transport** (pumping protons against their electrochemical concentration gradient).
Energy Source for Pumping	Exergonic electron transfer along the redox chain, derived from **electrons stored in NADH and $FADH_2$**.
Electron Route for NADH	Enters at **Complex I** → Ubiquinone (Q) → **Complex III** → Cytochrome c → **Complex IV** → O_2.
Electron Route for $FADH_2$	Enters at **Complex II** → Ubiquinone (Q) → **Complex III** → Cytochrome c → **Complex IV** → O_2.
Final Electron Acceptor	**Molecular Oxygen (O_2)**; reduced along with matrix protons to form H_2O.
Power Source for ATP Synthase	The **proton-motive force**: H^+ ions flowing down their electrochemical gradient from the intermembrane space back into the matrix.

Problem 4: Cellular Poisons Diagnostic Analysis (Active Learning #3, Slide 7)

Poison	Molecular Mechanism	Diagnostic Physiological Symptom & Rationale
Shikonin	Inhibits primary breakdown of glucose	**Extremely high intracellular glucose** with negligible ATP; glucose cannot enter glycolysis.
Arsenic	Inhibits pyruvate dehydrogenase (pyruvate → Acetyl-CoA)	**Massive pyruvate accumulation** with normal glucose; Citric Acid Cycle starved of Acetyl-CoA.
Rotenone	Blocks electron transfer from Complex I to Ubiquinone	**Elevated NADH levels**; $FADH_2$ consumption remains intact via Complex II; ATP production is moderately reduced.
DNP	Uncouples inner membrane by making it leaky to H^+	**Collapses proton gradient** (H^+ leaks into matrix); electron transport accelerates and releases heat without synthesizing ATP.
Oligomycin	Inhibits the F_0 rotor channel of ATP Synthase	**Extreme H^+ accumulation in intermembrane space** with completely arrested ATP synthesis; protons cannot re-enter matrix.